

Brain and Cognitive Science Club, IIT Kanpur

The Shelby Trap

Stage 4 Analysis Report

Q-Learning, Model-Free Blindspots, and the Daw Two-Step Task

Background: Q-Learning

Q-Learning is a **model-free** (this basically means that it does not plan the action in advance before doing it so it only knows about actions and doesn't monitor the transition probabilities), **off-policy** (which means it interacts and learns which policy to use that is mathematically represented by max in the equation) reinforcement learning algorithm. The agent learns a function $Q(s, a)$ representing the expected cumulative (mathematically by summation in expected return (G_t)) discounted (represented in equation by γ the discount factor) reward obtained by taking action a in state s and following the optimal policy thereafter.

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \quad (1)$$

The Bellman Update Equation

After every transition (s, a, r, s') , the agent updates its Q-table as:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a) \right] \quad (2)$$

where:

- $\alpha \in (0, 1]$ is the **learning rate** — how aggressively the agent updates its estimates
- $\gamma \in [0, 1]$ is the **discount factor** — how much future rewards are valued
- r is the **immediate reward** received
- s' is the **next state** actually observed
- $\max_{a'} Q(s', a')$ is the **best expected value** from the next state

The term in brackets is called the **TD error** (Temporal Difference error) — the gap between what the agent expected and what actually happened. The agent moves its Q-value a small step α in the direction of this error.

Exploration vs. Exploitation

The agent uses an ϵ -**greedy** policy: with probability ϵ it takes a random action (explore), and with probability $1 - \epsilon$ it takes the action with the highest Q-value (exploit).

Key Property: Model-Free

Q-Learning is *model-free*, meaning it does not maintain or use any model of the environment's transition dynamics $P(s' | s, a)$. It learns purely from sampled experience. As will be shown below, this is both its strength and its fundamental weakness.

The Two-Step Environment

The experiment uses a minimal Markov Decision Process with three states and two steps per episode:

- **State 0 (Hub):** The agent chooses A_1 or A_2 .
 - A_1 : leads to S_1 with probability 0.8 (Common), S_2 with probability 0.2 (Rare)
 - A_2 : leads to S_2 with probability 0.8 (Common), S_1 with probability 0.2 (Rare)
- **State 1 (Terminal):** reward $R = 1$ with probability 0.8
- **State 2 (Terminal):** reward $R = 1$ with probability 0.2

The optimal policy is clearly A_1 , since it reliably leads to the high-reward S_1 . The expected reward per episode under A_1 is:

$$\mathbb{E}[R | A_1] = 0.8 \times 0.8 + 0.2 \times 0.2 = 0.64 + 0.04 = 0.68$$

Question 1: The Mathematical Blindspot

Why is the Model-Free Agent Blind?

Consider the **forced trajectory** from Stage 3:

$$S_0 \xrightarrow{A_1} S_2 \text{ (Rare, 20\% transition)} \rightarrow R = 1$$

The Bellman update applied to $Q(S_0, A_1)$ is:

$$Q(S_0, A_1) \leftarrow Q(S_0, A_1) + \alpha \left[R + \gamma \cdot V(S_2) - Q(S_0, A_1) \right] \quad (3)$$

Since $R = 1 > 0$ and $\gamma \cdot V(S_2) > 0$, the TD error is positive and $Q(S_0, A_1)$ **increases** — the agent becomes more confident in A_1 .

The Specific Term that Causes Incorrect Credit

The blindspot lies in the **next state** s' used in the update. Equation 2 uses the *observed* $s' = S_2$ directly, with no weighting by how likely that transition was.

Crucially, **the transition probability** $P(S_2 | S_0, A_1) = 0.2$ **appears nowhere in the Bellman equation.** The update is identical regardless of whether the agent ended up in S_2 because of a 1% fluke or an 80% common outcome.

The agent sees: *“I took A_1 , I got a reward, therefore A_1 is better.”* It cannot reason: *“I took A_1 but ended up in S_2 , which only happens 20% of the time — this experience is not representative of what A_1 normally leads to.”*

This is **temporal credit assignment without structural awareness**: Q-Learning assigns credit to the action taken, not to the action that would reliably reproduce the outcome.

Question 2: The Model-Based Formulation

What a Model-Based Agent Must Store

A model-based agent maintains an explicit model of the environment:

- **Transition model:** $P(s' | s, a)$ — probability of each next state given state and action
- **Reward model:** $R(s')$ — expected reward from each state

Computing $V(S_0)$ with Transition Probabilities

The model-based agent computes the value of each action at S_0 by explicitly summing over all possible next states, weighted by their transition probabilities:

$$Q_{\text{MB}}(S_0, a) = \sum_{s'} P(s' | S_0, a) \cdot [R(s') + \gamma \cdot V(s')] \quad (4)$$

Expanding for both actions:

$$\begin{aligned} Q_{\text{MB}}(S_0, A_1) &= P(S_1 | S_0, A_1) \cdot R(S_1) + P(S_2 | S_0, A_1) \cdot R(S_2) \\ &= 0.8 \times 0.8 + 0.2 \times 0.2 = 0.68 \end{aligned} \quad (5)$$

$$\begin{aligned} Q_{\text{MB}}(S_0, A_2) &= P(S_2 | S_0, A_2) \cdot R(S_2) + P(S_1 | S_0, A_2) \cdot R(S_1) \\ &= 0.8 \times 0.2 + 0.2 \times 0.8 = 0.32 \end{aligned} \quad (6)$$

The model-based agent correctly selects A_1 ($0.68 > 0.32$). More importantly, after the forced rare trajectory, it would **not** update its preference for A_1 because it knows $P(S_2 | S_0, A_1) = 0.2$ and one rare sample does not change the stored transition model.

Question 3: The Hybrid Biological System

Neuroscientific Motivation

Biological brains are known to employ both systems simultaneously. Neuroscience identifies:

- **Striatum** (basal ganglia): implements model-free cached values, fast and habitual
- **Prefrontal cortex**: implements model-based planning, slower but flexible

The balance between these systems shifts depending on cognitive load, novelty, and uncertainty.

The Hybrid Decision Variable

A simple linear blend of the two systems yields the following decision variable:

$$Q_{\text{hybrid}}(s, a) = \lambda \cdot Q_{\text{MB}}(s, a) + (1 - \lambda) \cdot Q_{\text{MF}}(s, a) \quad (7)$$

where:

- $Q_{\text{MF}}(s, a)$ is the standard tabular Q-value (model-free)
- $Q_{\text{MB}}(s, a) = \sum_{s'} P(s' | s, a) \cdot [R(s') + \gamma \cdot V(s')]$ (model-based)
- $\lambda \in [0, 1]$ is the mixing weight

Adaptive Mixing

In biological systems λ is not fixed — it adapts based on cognitive state. A natural formulation uses uncertainty as the driver:

$$\lambda(s, a) = \sigma\left(\frac{1}{N(s, a)}\right) \quad (8)$$

where $N(s, a)$ is the visit count and σ is the sigmoid function. The intuition:

- When $N(s, a)$ is **small** (novel situation) $\rightarrow \lambda \rightarrow 1$: rely on the model, plan carefully
- When $N(s, a)$ is **large** (well-practised) $\rightarrow \lambda \rightarrow 0$: rely on cached Q-values, act fast

This mirrors the cognitive shift from deliberate planning (learning a new skill) to automaticity (expert performance) observed in human and animal behaviour.

Behaviour After the Forced Trajectory

Under the hybrid formulation, after the rare $A_1 \rightarrow S_2 \rightarrow R=1$ trajectory:

- $Q_{\text{MF}}(S_0, A_1)$ increases slightly (model-free credit assignment)
- $Q_{\text{MB}}(S_0, A_1)$ is unchanged at 0.68 (model knows the rare transition is rare)
- Q_{hybrid} moves only weakly, weighted by λ

The model-based component acts as a **regulariser**, preventing the agent from being misled by statistically unrepresentative experiences — exactly the behaviour a rational agent (or a human) would exhibit.

Summary

Property	Model-Free (Q-Learning)	Model-Based
Stores	Q-table $Q(s, a)$	Transition model $P(s' s, a)$
Update	Sampled TD error	Full expectation over s'
Rare transition	Incorrectly credits action	Correctly discounts by P
Speed	Fast, low memory	Slower, high memory
Biological analog	Striatum / habit system	Prefrontal cortex / planning

The Shelby Trap demonstrates that purely model-free agents, like Leonard Shelby relying only on his tattoos, are fundamentally limited by their inability to reason about the *structure* of the world. A hybrid system that blends model-free efficiency with model-based structural reasoning is both computationally motivated and biologically plausible.